

Faculty of Mathematics and Computer Science

Machine learning course (ML)

Support Vector Machine Approaches to Human Activity Recognition Using Smartphone Sensors

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Topic description and relevance

Human activity recognition (HAR) is the task of automatically identifying what a person is doing, such as walking, climbing stairs, sitting or lying down, from the signals of sensors carried or worn by that person. Modern smartphones embed a triaxial accelerometer and gyroscope, so they can act as unobtrusive, low-cost activity monitors without any extra hardware. The usual pipeline splits the raw inertial signals into short windows, computes time- and frequency-domain features for every window and feeds the resulting feature vectors to a supervised classifier. The public UCI HAR dataset, recorded from 30 subjects carrying a waist-mounted smartphone and described by 561 such features, has become the reference benchmark for this pipeline [3].

The topic is relevant from a practical perspective because activity information underpins many applications: ambient assisted living and remote monitoring of elderly people, fall and inactivity detection, rehabilitation and chronic disease management, fitness tracking and context-aware mobile services. These applications run on battery-powered devices with limited computing resources, so the classifier has to be accurate, robust to differences between users and phone placements, and cheap enough to run on the device itself [2, 6].

From a theoretical perspective, support vector machines (SVMs) are a natural choice for this problem. An SVM learns the decision boundary with the largest margin between classes and, through kernel functions, can separate data that is not linearly separable in the original feature space [5]. Margin maximisation gives good generalisation on high-dimensional feature vectors such as the 561 HAR features, and multiclass problems can be handled with one-vs-one or one-vs-all decompositions. A recent survey reviews these properties together with the open challenges of SVM classification, including multiclass extensions, large and unbalanced datasets and the selection of kernel hyperparameters [4].

SVMs have played a central role in smartphone HAR research. The authors of the UCI HAR dataset proposed a hardware-friendly multiclass SVM that uses fixed-point arithmetic to reduce the computational cost on the phone while keeping accuracy close to that of a standard SVM [2], and used a multiclass SVM to report the reference results

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on the dataset [3]. Later work extended the approach to postural transitions and unknown activities by combining the probabilistic outputs of consecutive SVM predictions with heuristic filtering [8]. More recent studies improve SVM-based HAR through feature selection, using hybrid filter–wrapper schemes [1] or metaheuristic optimisers [7], and evaluate SVMs on real-life data collected with free phone orientation and placement [6]. Deep learning models such as convolutional networks have since been proposed for the same task and are usually compared against SVM baselines [9].

The report will survey these approaches, compare their methodology, datasets and experimental results, and critically discuss the trade-offs between accuracy, computational cost and robustness of SVM-based HAR, including how SVMs compare with deep learning alternatives. It will also motivate the software project, in which a multiclass SVM will be implemented from scratch and with an existing library and evaluated on the UCI HAR dataset.

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